

A Modular and Configurable Architecture for ROS 2 Hardware Integration with micro-ROS

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ABSTRACT

Integrating resource-constrained hardware into ROS 2 is complex. This thesis presents a modular and configurable architecture utilizing micro-ROS to interface embedded microcontrollers with a central ROS 2 ecosystem, detailing network synchronization, payload limits, and memory management under real-time operating systems.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Guides the software integration framework for communicating the ESP32 S3 with high-level ROS 2 nodes or supervisors.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Jakob Friedl provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.